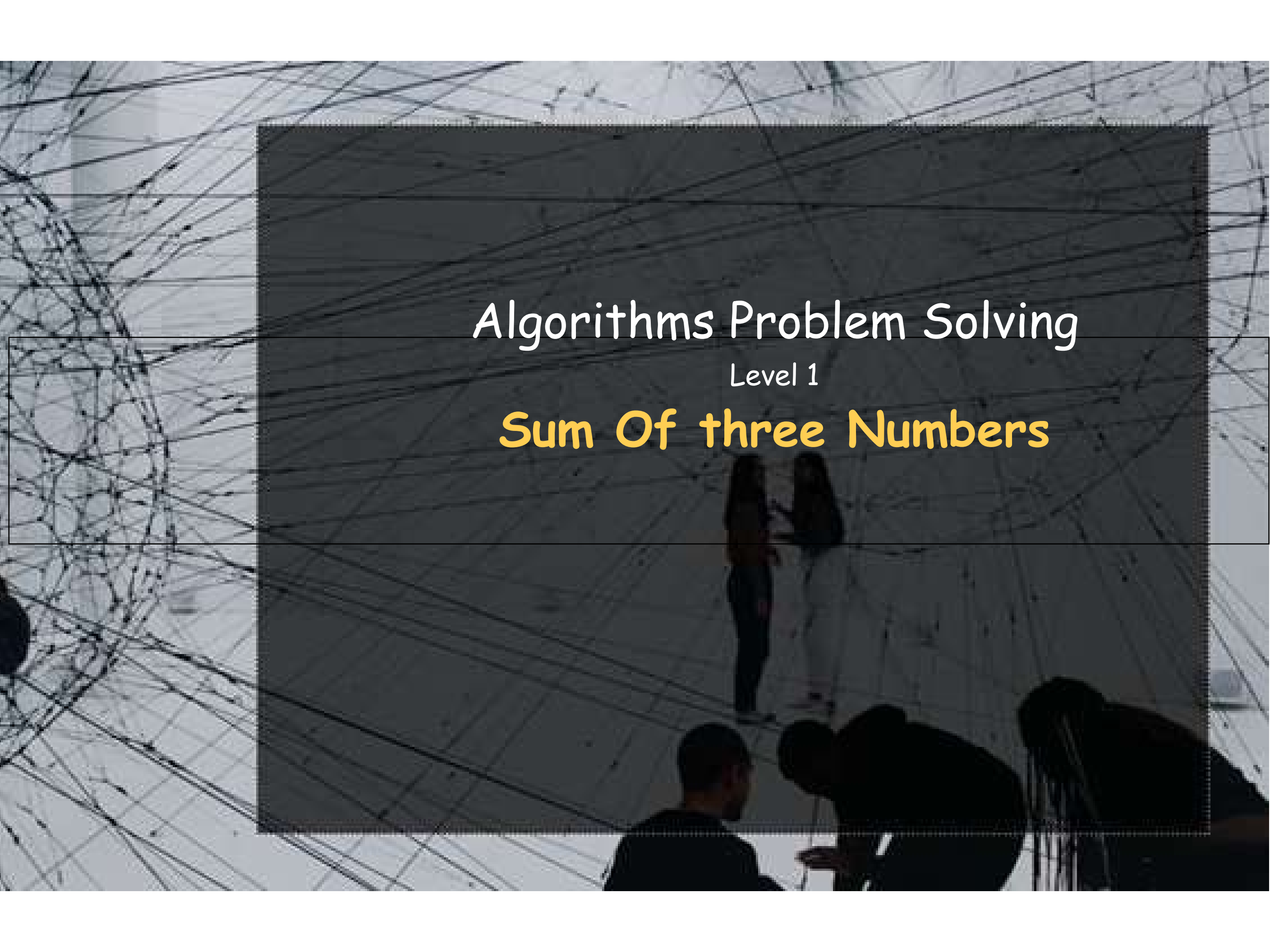


Algorithms Problem Solving Level 1

Sum Of three Numbers



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Definition

- ◆ This program reads three numbers from the user and calculates their sum.
- ◆ The program uses a structure to group the three numbers and separate functions to read, calculate, and print the result.
- ◆ The sum is calculated by adding the three structure members together.

Explanation

- ◆ The struct `stNumbers` groups three integer values:
 - Num1 stores the first number.
 - Num2 stores the second number.
 - Num3 stores the third number.
- ◆ `ReadNumbers()` reads the three numbers from the user and returns an `stNumbers` object.
- ◆ `sumNumbers()` receives the structure and adds Num1, Num2, and Num3.
- ◆ `PrintResult()` receives the structure and calls `sumNumbers()` to calculate and print the result.
- ◆ In `main()`, the functions are combined:
`PrintResult(ReadNumbers());`

Steps in main()

1. Create the stNumbers structure.
2. Read the first number.
3. Read the second number.
4. Read the third number.
5. Return the structure containing the three numbers.
6. Pass the structure to PrintResult().
7. Call sumNumbers() to calculate the sum.
8. Print the calculated result.

💡 Example

◆ If the user enters:

First number: 10

Second number: 20

Third number: 30

◆ The calculation is:

$$10 + 20 + 30 = 60$$

➡ Result

The sum of the three numbers is: 60

You have failed.

Note

- ◆ A structure allows the three related numbers to be grouped into one object.
- ◆ The same `stNumbers` object can be passed between different functions.
- ◆ `sumNumbers()` returns an `int` because all three numbers are integers.

⊘ Warning

- ◆ Make sure all three numbers are read before calculating the sum.
- ◆ The program does not validate whether the user enters valid integer input.



Comparison

▨ Using a Structure

- ✓ Groups the three related numbers into one object.
- ✓ Makes it easier to pass all three numbers to functions.
- ✓ Accesses the values using Num1, Num2, and Num3.

▤ Using Separate Variables

- ✓ Each number would be stored in a separate variable.
- ✓ All three variables would need to be passed separately to functions.
- ✓ Can become less convenient when working with more related data.

✦✦ Summary

- ✓ A structure can group multiple related numbers into one object.
- ✓ Functions can use the structure to read, calculate, and print the result.
- ✓ The program calculates the sum by adding the three structure members together.

Thank you 😊